

The Six-Point Type-II Bounded Ratio on Its Symmetric Locus, and Two Exact Obstructions

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Abstract

For the normalized six-point type-II hypermetric vector $h = (-2, -1, 1, 1, 1, 1)$, we determine the sharp supremum of the associated bounded ratio on the complete locus fixed by permutations of the last four indices. The value is $8/(3\sqrt{3})$, strictly below 2. We then give two exact obstructions to extending this calculation by natural reductions: entrywise geometric orbit averaging can create a second positive eigenvalue, and the target exponent is separated from the product cone generated by all lifted triangular and pentagonal inequalities and the elementary inequalities $p_{ij}^{-1} \leq 1$. These are restricted and proof-method results. They do not bound arbitrary nonsymmetric type-II matrices, treat the other genuinely six-point type, or resolve the original six-point bound and equality classification.

1 Context and statement of the partial result

Let

$$\mathcal{L}_{6,1}^+ = \{P = (p_{ij}) \in \mathbb{R}^{6 \times 6} : P = P^\top, p_{ii} = 1, p_{ij} > 0 \ (i \neq j), \\ P \text{ has at most one positive eigenvalue}\}.$$

For $h \in \mathbb{Z}^6$ put $q(h)_{ij} = h_i h_j$ for $i < j$. We study

$$h_* = (-2, -1, 1, 1, 1, 1), \quad \alpha_* = \frac{q(h_*)}{4}, \quad F_*(P) = \prod_{i < j} p_{ij}^{(\alpha_*)_{ij}}. \quad (1.1)$$

Since $\sum_i h_{*,i} = 1$ and $\sum_i h_{*,i}^2 = 9$, we have $\sum_{i < j} (\alpha_*)_{ij} = -1$, so this is the normalized type-II six-point exponent.

Huang, Huh, Soskin, and Wang identify the cone of reduced bounded ratios for Lorentzian matrices with the negative dual of the cut cone and formulate the uniform normalized bound by 2 [2]. Baldi and Kummer give a seven-point counterexample arising from a non-hypermetric clique-web facet; for six points they record two additional hypermetric facet types beyond lifts from five points and report that their search did not produce a counterexample for those types [1]. That report is not a proof of the six-point assertion. The present note concerns only the type-II vector in (1.1), and even there only a symmetric subfamily and two specified proof mechanisms.

Write $G = S_{\{3,4,5,6\}}$. The three exact conclusions are as follows.

Theorem 1.1. *For α_* in (1.1):*

1. *On the full fixed locus $(\mathcal{L}_{6,1}^+)^G$,*

$$\sup_{P \in (\mathcal{L}_{6,1}^+)^G} F_*(P) = \frac{8}{3\sqrt{3}} < 2. \quad (1.2)$$

2. Entrywise geometric averaging over G does not preserve $\mathcal{L}_{6,1}^+$. More precisely, the family P_r in Proposition 3.1 has inertia $(1, 1, 4)$ for every $r > 1$, whereas its geometric orbit average has inertia $(2, 4, 0)$. The averaging preserves F_* , and both values equal 1.
3. The exponent α_* is not in the cone generated by all lifted triangular exponents q_T , all normalized lifted pentagonal exponents $q_P/2$, and the elementary exponents $-e_{ij}$. Consequently $F_*(P) \leq 2$ cannot be obtained by multiplying nonnegative real powers of those known inequalities, even after removing the usual total-weight cap.

The next three sections prove the three parts separately.

2 Sharp optimization on the fixed locus

Every G -fixed matrix has the form

$$P(a, b, c, d) = \begin{pmatrix} 1 & a & b & b & b & b \\ a & 1 & c & c & c & c \\ b & c & 1 & d & d & d \\ b & c & d & 1 & d & d \\ b & c & d & d & 1 & d \\ b & c & d & d & d & 1 \end{pmatrix}, \quad a, b, c, d > 0. \quad (2.1)$$

Put $s = 1 + 3d$. On the three-dimensional contrast space supported on the last four coordinates, P acts as $(1 - d)I_3$. On the orthogonal complement, in a normalized basis, it is represented by

$$Q = \begin{pmatrix} 1 & a & 2b \\ a & 1 & 2c \\ 2b & 2c & s \end{pmatrix}, \quad \det Q = (1 - a^2)(s - 4b^2) - 4(c - ab)^2. \quad (2.2)$$

Lemma 2.1. *The matrix in (2.1) belongs to $\mathcal{L}_{6,1}^+$ if and only if*

$$d \geq 1, \quad a \geq 1, \quad 4b^2 \geq s, \quad (1 - a^2)(s - 4b^2) \geq 4(c - ab)^2. \quad (2.3)$$

Proof. If P has at most one positive eigenvalue, the contrast eigenvalues give $d \geq 1$. The 2×2 compression on the first two coordinates and the corresponding 2×2 compression inside Q give respectively $a \geq 1$ and $4b^2 \geq s$. The block Q has one positive eigenvalue and two nonpositive eigenvalues, so $\det Q \geq 0$, which is the last condition.

Conversely, suppose (2.3) holds. If $a > 1$, the leading 2×2 block of Q has one positive and one negative eigenvalue. Interlacing makes the largest eigenvalue of Q positive and the smallest negative. Since $\det Q \geq 0$, its middle eigenvalue is nonpositive. If $a = 1$, the final condition in (2.3) forces $c = b$. Then $(1, -1, 0)$ is a null vector of Q , and the remaining 2×2 block has determinant $2(s - 4b^2) \leq 0$. Thus Q has at most one positive eigenvalue in every case. Together with $1 - d \leq 0$ on the contrast space, this proves the claim, including the singular boundary cases. $\square$

For (2.1), direct collection of the exponents in (1.1) gives

$$F_*(P) = \frac{\sqrt{a} d^{3/2}}{b^2 c}. \quad (2.4)$$

Proposition 2.2. *The supremum in (1.2) is $8/(3\sqrt{3})$, and it is approached by an explicit one-parameter family.*

Proof. Under (2.3), write uniquely

$$a = \cosh x, \quad \frac{2b}{\sqrt{s}} = \cosh y, \quad x, y \geq 0.$$

The determinant condition in (2.3) implies

$$\begin{aligned} c &\geq ab - \frac{1}{2} \sqrt{(a^2 - 1)(4b^2 - s)} \\ &= \frac{\sqrt{s}}{2} \cosh(x - y). \end{aligned} \tag{2.5}$$

Substitution into (2.4) yields

$$F_*(P) \leq 8 \left(\frac{d}{s} \right)^{3/2} \frac{\sqrt{\cosh x}}{\cosh^2 y \cosh(x - y)}. \tag{2.6}$$

The last factor is at most one. To see this, set $z = x - y$, $p = \tanh y \geq 0$, and $q = \tanh z$. After squaring and using $\cosh x = \cosh y \cosh z(1 + pq)$, the desired estimate becomes

$$1 + pq \leq \cosh^3 y \cosh z. \tag{2.7}$$

If $q \leq 0$, the left side is at most 1 and the right side is at least 1. If $q \geq 0$, put $t = \sqrt{(1 - p^2)(1 - q^2)}$. Cauchy-Schwarz gives

$$t + pq \leq 1.$$

Since $0 < t \leq 1$ and $pq \geq 0$, we have $t(1 + pq) \leq t + pq \leq 1$. Thus $1 + pq \leq t^{-1} = \cosh y \cosh z$, which is stronger than (2.7). Therefore

$$F_*(P) \leq 8 \left(\frac{d}{1 + 3d} \right)^{3/2} < \frac{8}{3\sqrt{3}}. \tag{2.8}$$

For every $d \geq 1$, take

$$a = 1, \quad b = c = \frac{\sqrt{1 + 3d}}{2}. \tag{2.9}$$

These parameters satisfy (2.3) and give equality in the first bound of (2.8). Letting $d \rightarrow \infty$ proves that the supremum is exactly $8/(3\sqrt{3})$. It is a supremum rather than a claimed finite- d maximum. $\square$

3 Failure of geometric orbit symmetrization

For a positive matrix P , define its entrywise geometric orbit average by

$$\mathcal{G}(P)_{ij} = \exp \left(\frac{1}{|G|} \sum_{g \in G} \log(g \cdot P)_{ij} \right). \tag{3.1}$$

Because α_* is G -invariant, $F_*(\mathcal{G}(P)) = F_*(P)$. The following proposition shows that this identity cannot be combined with closure of the Lorentzian domain.

Proposition 3.1. For $r > 1$, split the indices into $A = \{1, 5, 6\}$ and $B = \{2, 3, 4\}$, and set

$$(P_r)_{ii} = 1, \quad (P_r)_{ij} = \begin{cases} 1, & i, j \text{ lie in the same part,} \\ r, & i, j \text{ lie in different parts.} \end{cases} \quad (3.2)$$

Then $\text{In}(P_r) = (1, 1, 4)$, while $\text{In}(\mathcal{G}(P_r)) = (2, 4, 0)$. Moreover, $F_*(P_r) = F_*(\mathcal{G}(P_r)) = 1$.

Proof. On the four-dimensional space of vectors whose coordinate sums on A and B are both zero, P_r vanishes. On the two part-constant unit vectors its matrix is

$$\begin{pmatrix} 3 & 3r \\ 3r & 3 \end{pmatrix},$$

whose eigenvalues are $3(1+r) > 0$ and $3(1-r) < 0$. This proves the first inertia statement.

Orbit counting in (3.1) puts $\mathcal{G}(P_r)$ in the form (2.1) with

$$a = r, \quad b = c = \sqrt{r}, \quad d = r^{2/3}. \quad (3.3)$$

Its three last-coordinate contrasts have eigenvalue $1 - r^{2/3} < 0$. In the block Q from (2.2), $(1, -1, 0)$ has eigenvalue $1 - r < 0$. On the orthogonal complement of that vector, the remaining block is

$$B_r = \begin{pmatrix} 1+r & 2\sqrt{2r} \\ 2\sqrt{2r} & 1+3r^{2/3} \end{pmatrix}.$$

Writing $x = r^{1/3} > 1$, we have

$$\det B_r = (x-1)^2(3x^3 + 6x^2 + 2x + 1) > 0. \quad (3.4)$$

Both diagonal entries are positive, so B_r is positive definite and the second inertia statement follows.

Finally,

$$\sum_{i \in A, j \in B} (\alpha_*)_{ij} = \frac{1}{4} \left(\sum_{i \in A} h_{*,i} \right) \left(\sum_{j \in B} h_{*,j} \right) = 0,$$

because the first factor is zero. All non-unit entries of P_r are cross entries, so $F_*(P_r) = 1$, and (3.1) preserves this value. $\square$

For a fully explicit witness, $r = 8$ gives nonzero spectrum $\{27, -21\}$ for P_8 . The geometric average has spectrum

$$\{11 + 2\sqrt{17}, 11 - 2\sqrt{17}, -7, -3, -3, -3\}.$$

Thus the failure of closure is certified without numerical approximation.

4 Separation from the lifted product-proof cone

For any $P \in \mathcal{L}_{6,1}^+$, interlacing applied to the principal block $\begin{pmatrix} 1 & p_{ij} \\ p_{ij} & 1 \end{pmatrix}$ shows that $p_{ij} \geq 1$. Hence $p_{ij}^{-1} \leq 1$ is always available.

Let q_T run over the 60 distinct permutations of $q(1, 1, -1, 0, 0, 0)$, and let q_P run over the 60 distinct permutations of $q(1, 1, 1, -1, -1, 0)$. These are the lifted triangular exponents and the unnormalized lifted pentagonal exponents, respectively. Define

$$\mathcal{C} = \text{cone}(\{q_T\}_T \cup \{q_P/2\}_P \cup \{-e_{ij} : i < j\}). \quad (4.1)$$

A usual direct product certificate for $F_*(P) \leq 2$ would in particular place α_* in this cone; the additional requirement that the total triangular and pentagonal weight be at most one only controls the resulting constant. We exclude even the uncapped conic representation.

Proposition 4.1. *The functional*

$$\Lambda(v) = 3v_{12} + \sum_{k=3}^6 v_{1k} + 2 \sum_{k=3}^6 v_{2k} + 2 \sum_{3 \leq i < j \leq 6} v_{ij} \quad (4.2)$$

separates α_ from $\mathcal{C}$. In particular, $\alpha_* \notin \mathcal{C}$.*

Proof. All coefficients in (4.2) are positive, so $\Lambda(-e_{ij}) < 0$. Direct substitution gives

$$\Lambda(q(h_*)) = 3(2) + 4(-2) + 2 \cdot 4(-1) + 2 \cdot 6(1) = 2, \quad \Lambda(\alpha_*) = \frac{1}{2} > 0. \quad (4.3)$$

For completeness, exact enumeration of the two finite lifted families gives the following complete histograms:

$$\begin{array}{c|ccc} \Lambda(q_T) & -4 & -2 & 0 \\ \hline \text{number of lifts} & 4 & 46 & 10 \end{array} \quad \begin{array}{c|ccc} \Lambda(q_P) & -6 & -4 & -2 & 0 \\ \hline \text{number of lifts} & 12 & 28 & 16 & 4. \end{array} \quad (4.4)$$

These counts can be checked directly from

$$\Lambda(q(h)) = 3h_1h_2 + h_1 \sum_{k=3}^6 h_k + 2h_2 \sum_{k=3}^6 h_k + 2 \sum_{3 \leq i < j \leq 6} h_ih_j$$

by choosing the support and sign locations: there are $\binom{6}{3} \cdot 3 = 60$ triangular choices and $6\binom{5}{2} = 60$ pentagonal choices. Thus every generator of $\mathcal{C}$ has nonpositive Λ -value, whereas (4.3) is positive. Linearity proves the separation. $\square$

The conclusion concerns precisely the cone (4.1). It does not rule out a proof using sums, auxiliary inequalities, nonlinear elimination, or a direct analysis of the full semialgebraic matrix domain.

5 Scope, methodology, and reproducibility

The exact fixed-locus calculation and the two obstructions form a partial result, not a resolution of the six-point problem. Arbitrary nonsymmetric type-II matrices remain untreated; so does the other genuinely six-point hypermetric type. Consequently this note neither proves nor disproves the uniform six-point bound and does not classify equality extreme rays.

The proofs above use symbolic factorizations, elementary inertia arguments, and a finite integer enumeration. A companion verification script implements the 120 evaluations in (4.4) using only combinatorial generation and integer arithmetic; no floating-point evidence is used in any theorem. Exact rational, algebraic, and Bernstein certificates on explicitly delimited regions are also emphasized by Zeng, Liu, Ratnavelu, Huat, and Xiong in a different Hermitian scalar-gate problem [3]. That paper does not concern Lorentzian bounded ratios or cut cones; the comparison is methodological only.

A language model assisted with proof exploration and manuscript preparation. The displayed determinant identities, inertia arguments, inequalities, and finite separator calculation were checked from their definitions with exact arithmetic. No formal proof assistant was used. The literature comparison is bounded to the cited sources and is not a claim of priority.

References

- [1] Lorenzo Baldi and Mario Kummer. A note on bounded ratios, 2026. URL: <https://arxiv.org/abs/2609.03934>, arXiv:2609.03934, doi:10.48550/arXiv.2609.03934.
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